

GROUP ASSIGNMENT – MINI PROJECT

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TechStart Solutions Linux Server Implementation

Professional System Documentation

1. Executive Summary

Project Overview

This project involved deploying a secure, multi-user Linux environment on AWS for **TechStart Solutions**, supporting 10 development team members across backend, frontend, and DevOps roles. The system was built with security, collaboration, and administrative efficiency as core principles.

AWS Infrastructure Summary

- **Instance Type:** t3.micro (Free Tier eligible)
- **Operating System:** Ubuntu Server 22.04 LTS
- **Storage:** 8 GB EBS Volume
- **Network:** Secure SSH-only access with key-based authentication
- **Region:** Europe(Stockholm)

Total Users and Groups Created

- **Users:** 10 (3 Backend, 3 Frontend, 3 DevOps, 1 Team Lead)
- **Groups:** 5 (backend-dev, frontend-dev, devops-team, dev-all, team-leads)

Key Security Measures Implemented

- Principle of Least Privilege enforcement
- SGID-enabled collaborative directories
- Tiered sudo access controls
- Password aging policies (90-day maximum)
- Secure SSH key authentication

- Regular backup procedures

2. AWS Infrastructure Documentation

Instance Details

- **Instance ID:** i-0abcdef1234567890
- **Instance Type:** t3.micro
- **AMI:** ami-0abcdef1234567890 (Ubuntu 22.04 LTS)

Network Configuration

- **VPC:** vpc-0abcdef1234567890 (Default)
- **Subnet:** subnet-0abcdef1234567890
- **Public IP:** 13.49.64.209
- **Private IP:** 172.31.16.184

Phase 1: Building the Foundation

1. Launching the Instance:

1. **Review all your settings:**
 - Instance type: t3.micro
 - Storage: 8 GB
 - Security group: SSH only from My IP
 - Key pair: Your Group4-TechStart-Key.pem

Instances (1/1) Info

Find Instance by attribute or tag (case-sensitive) All states

Instance state = running Clear filters

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability Zone
Group4-TechStart-Server	i-02f06c46a83af8c5d	Running	t3.micro	3/3 checks passed	View alarms	eu-north-1a

i-02f06c46a83af8c5d (Group4-TechStart-Server)

Details Status and alarms Monitoring Security Networking Storage Tags

Instance summary Info

Instance ID i-02f06c46a83af8c5d	Public IPv4 address 13.49.64.209 open address	Private IPv4 addresses 172.31.16.184
IPv6 address -	Instance state Running	Public DNS ec2-13-49-64-209.eu-north-1.compute.amazonaws.com

i-02f06c46a83af8c5d (Group4-TechStart-Server)

Answer private resource DNS name IPv4 (A)	Instance type t3.micro	Elastic IP addresses -
Auto-assigned IP address 13.49.64.209 [Public IP]	VPC ID vpc-0ddd416fea11eb4cf	AWS Compute Optimizer finding Opt-In to AWS Compute Optimizer for recommendation s. Learn more
IAM Role -	Subnet ID subnet-0e565061c2e79809e	Auto Scaling Group name -

2. Connecting AWS with the ubuntu

Command used: `ssh -i /sano/Downloads/Group4-TechStart-Key.pem ubuntu@13.59.64.209`

```
sano@sano-VirtualBox:~$ pwd
/home/sano
sano@sano-VirtualBox:~$ ssh -i /Users/yourname/Downloads/Group5-TechStart-Key.pem ubuntu@54.210.123.45
Warning: Identity file /Users/yourname/Downloads/Group5-TechStart-Key.pem not accessible: No such file or directory.

zssh: connect to host 54.210.123.45 port 22: Connection refused
sano@sano-VirtualBox:~$ 
sano@sano-VirtualBox:~$ pwd
/home/sano
sano@sano-VirtualBox:~$ ls
Desktop    Downloads      Music      Public      snap      Videos
Documents  Introduction_to_linux  Pictures  PycharmProjects  Templates
sano@sano-VirtualBox:~$ cd D
Desktop/  Documents/ Downloads/
sano@sano-VirtualBox:~$ cd Downloads/
sano@sano-VirtualBox:~/Downloads$ ls
custom_training_module      'Delivery Slip - Wood Corner - WH_IN_00005.pdf'
'custom_training_module (2)' 'Delivery Slip - Wood Corner - WH_OUT_00001.pdf'
'custom_training_module (3)' sale_order.cpython-310.pyc
custom_training_module.zip
sano@sano-VirtualBox:~/Downloads$ ssh -i /sano/Downloads/Group4-TechStart-Key.pem ubuntu@13.49.64.209
```

3. Created User Accounts for each group member

Names:

- sano_kabayiza_jules
- iradukunda_oscar
- ishimwe_amos

```
ubuntu@ip-172-31-16-184:~$ ls
ubuntu@ip-172-31-16-184:~$ sudo adduser sano_kabayiza_jules
Adding user `sano_kabayiza_jules' ...
Adding new group `sano_kabayiza_jules' (1001) ...
Adding new user `sano_kabayiza_jules' (1001) with group `sano_kabayiza_jules'
.
Creating home directory `/home/sano_kabayiza_jules' ...
Copying files from `/etc/skel' ...
New password:
Retype new password:
passwd: password updated successfully
Changing the user information for sano_kabayiza_jules
Enter the new value, or press ENTER for the default
  Full Name []: sano_kabayiza_jules
  Room Number []: 1
  Work Phone []: 0786992184
  Home Phone []:
  Other []:
Is the information correct? [Y/n] y
ubuntu@ip-172-31-16-184:~$ sudo adduser iradukunda_oscar
Adding user `iradukunda_oscar' ...
Adding new group `iradukunda_oscar' (1002) ...
Adding new user `iradukunda_oscar' (1002) with group `iradukunda_oscar' ...
Creating home directory `/home/iradukunda_oscar' ...
Copying files from `/etc/skel' ...
```

```
ubuntu@ip-172-31-16-184:~$ sudo usermod -aG sudo sano_kabayiza_jules
ubuntu@ip-172-31-16-184:~$ sudo usermod -aG sudo iradukunda_oscar
ubuntu@ip-172-31-16-184:~$ sudo usermod -aG sudo ishimwe_amos
ubuntu@ip-172-31-16-184:~$ sudo mkdir -p /opt/techstart/
ubuntu@ip-172-31-16-184:~$ client_loop: send disconnect: Broken pipe
sano@sano-VirtualBox:~/Downloads$ hostname
sano-VirtualBox
sano@sano-VirtualBox:~/Downloads$ ssh -i Group4-TechStart-Key.pem ubuntu@13.49.6
4.209
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1035-aws x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Oct  6 13:10:12 UTC 2025

System load:  0.0               Processes:    108
```

PART 2: (BY STUDENT 2): System Preparation

Commands to run:

- **hostname** → Your server's name (ip-172-31-16-184)
- **lsb_release -a** → Ubuntu 22.04.5 LTS
- **uname -r** → Linux kernel version 6.8.0-1039-aws
- **free -h** → Memory/RAM usage (shows 23% in your case)
- **df -h** → Disk space (shows 22.4% of 7.57GB used)
- **ip addr show** → Network IP addresses (172.31.16.184)
- **date** → Current date and time
- **ls -ld /opt/techstart/** → Confirms the directory exists

```

Last login: Mon Oct  6 12:48:10 2025 from 102.22.168.153
ubuntu@ip-172-31-16-184:~$ ls
ubuntu@ip-172-31-16-184:~$ hostname
ip-172-31-16-184
ubuntu@ip-172-31-16-184:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 22.04.5 LTS
Release:        22.04
Codename:       jammy
ubuntu@ip-172-31-16-184:~$ uname -r
6.8.0-1035-aws
ubuntu@ip-172-31-16-184:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           914Mi       226Mi       166Mi       2.0Mi       521Mi       503Mi
Swap:              0B           0B           0B
ubuntu@ip-172-31-16-184:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/root        7.6G  2.4G  5.3G  31% /
tmpfs            458M   0  458M   0% /dev/shm
tmpfs            183M  884K  182M   1% /run
tmpfs            5.0M   0   5.0M   0% /run/lock
efivarfs         128K  3.6K  120K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p15  105M  6.1M   99M   6% /boot/efi
efivarfs         128K  3.6K  120K   3% /sys/firmware/efi/efivars
/dev/nvme0n1p15  105M  6.1M   99M   6% /boot/efi
tmpfs            92M  4.0K   92M   1% /run/user/1000
ubuntu@ip-172-31-16-184:~$ ip addr show
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens5: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 9001 qdisc mq state UP group default qlen 1000
    link/ether 06:19:b3:8c:30:41 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    inet 172.31.16.184/20 metric 100 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 2165sec preferred_lft 2165sec
    inet6 fe80::419:b3ff:fe8c:3041/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ip-172-31-16-184:~$ date
Mon Oct  6 13:16:02 UTC 2025
ubuntu@ip-172-31-16-184:~$ ls -ld /opt/techstart/
drwxr-xr-x 2 root root 4096 Oct  6 13:01 /opt/techstart/
ubuntu@ip-172-31-16-184:~$ S

```

Part 2: Group Creation

- Created 5 groups for organizing users into teams
- Each group gets automatically assigned a GID (Group ID number)
- The verification command shows all groups with their GIDs

```

ult qlen 1000
    link/ether 06:19:b3:8c:30:41 brd ff:ff:ff:ff:ff:ff
    altname enp0s5
    inet 172.31.16.184/20 metric 100 brd 172.31.31.255 scope global dynamic ens5
        valid_lft 2165sec preferred_lft 2165sec
    inet6 fe80::419:b3ff:fe8c:3041/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ip-172-31-16-184:~$ date
Mon Oct  6 13:16:02 UTC 2025
ubuntu@ip-172-31-16-184:~$ ls -ld /opt/techstart/
drwxr-xr-x 2 root root 4096 Oct  6 13:01 /opt/techstart/
ubuntu@ip-172-31-16-184:~$ sudo groupadd backend-dev
ubuntu@ip-172-31-16-184:~$ sudo groupadd frontend-dev
ubuntu@ip-172-31-16-184:~$ sudo groupadd devops-team
ubuntu@ip-172-31-16-184:~$ sudo groupadd dev-all
ubuntu@ip-172-31-16-184:~$ sudo groupadd team-leads
ubuntu@ip-172-31-16-184:~$ getent group | grep -E "(backend-dev|frontend-dev|devops-team|dev-all|team-leads)"
backend-dev:x:1004:
frontend-dev:x:1005:
devops-team:x:1006:
dev-all:x:1007:
team-leads:x:1008:
ubuntu@ip-172-31-16-184:~$

```

Part 3: User Account Creation(STUDENT3)

Created 10 Users; 3 in Frontend Developers, 3 in Backend Developers, 3 in Devops Engineers and 1 Team lead.

The password is: 123 for each user.

```

ubuntu@ip-172-31-16-184:~$ sudo usermod -aG dev-all devops3
ubuntu@ip-172-31-16-184:~$ sudo useradd -m -s /bin/bash -c "Lamine Yamal" -g team-leads teamlead
ubuntu@ip-172-31-16-184:~$ sudo passwd teamlead
New password:
Retype new password:
passwd: password updated successfully
ubuntu@ip-172-31-16-184:~$ sudo usermod -aG dev-all,sudo teamlead
ubuntu@ip-172-31-16-184:~$ id backend1
uid=1004(backend1) gid=1004(backend-dev) groups=1004(backend-dev),1007(dev-all)
ubuntu@ip-172-31-16-184:~$ getent passwd | grep -E "(backend|frontend|devops|teamlead)"
backend1:x:1004:1004:Ishimwe Yves:/home/backend1:/bin/bash
backend2:x:1005:1004:Ngabo Prime:/home/backend2:/bin/bash
backend3:x:1006:1004:Shema Yvan:/home/backend3:/bin/bash
frontend1:x:1007:1005:Sarah Chen:/home/frontend1:/bin/bash
frontend2:x:1008:1005:Berwa Nelly:/home/frontend2:/bin/bash
frontend3:x:1009:1005:Mucyo Happy:/home/frontend3:/bin/bash
devops1:x:1010:1006:Michael Brown:/home/devops1:/bin/bash
devops2:x:1011:1006:Lebron James:/home/devops2:/bin/bash
devops3:x:1012:1006:Shai Alexander:/home/devops3:/bin/bash
teamlead:x:1013:1008:Lamine Yamal:/home/teamlead:/bin/bash
ubuntu@ip-172-31-16-184:~$ S

```

Part 4: Shared Directory Structure

Step 1: Create Directory Structure

Step 2: Set Ownership and Permissions

```
backend && sudo chmod 2775 /opt/techstart/projects/backend
ubuntu@ip-172-31-16-184:~$ sudo chown root:frontend-dev /opt/techstart/projects/
frontend && sudo chmod 2775 /opt/techstart/projects/frontend
ubuntu@ip-172-31-16-184:~$ sudo chown root:devops-team /opt/techstart/projects/i
nfrastructure && sudo chmod 2775 /opt/techstart/projects/infrastructure
ubuntu@ip-172-31-16-184:~$ sudo chown root:dev-all /opt/techstart/shared && sudo
chmod 2775 /opt/techstart/shared
ubuntu@ip-172-31-16-184:~$ sudo chown root:dev-all /opt/techstart/shared/docs &&
sudo chmod 2775 /opt/techstart/shared/docs
ubuntu@ip-172-31-16-184:~$ sudo chown root:dev-all /opt/techstart/shared/resourc
es && sudo chmod 2775 /opt/techstart/shared/resources
ubuntu@ip-172-31-16-184:~$ sudo chown root:team-leads /opt/techstart/logs && sud
o chmod 2770 /opt/techstart/logs
ubuntu@ip-172-31-16-184:~$ sudo chown root:root /opt/techstart/backups && sudo c
hmod 755 /opt/techstart/backups
ubuntu@ip-172-31-16-184:~$ sudo ls -la /opt/techstart/
total 24
drwxr-xr-x 6 root root      4096 Oct  6 14:28 .
drwxr-xr-x 3 root root      4096 Oct  6 13:01 ..
drwxr-xr-x 2 root root      4096 Oct  6 14:28 backups
drwxrws--- 2 root team-leads 4096 Oct  6 14:28 logs
drwxr-xr-x 5 root root      4096 Oct  6 14:28 projects
drwxrwsr-x 4 root dev-all   4096 Oct  6 14:28 shared
ubuntu@ip-172-31-16-184:~$ S
```

Complete directory structure

```
no VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-16-184:~$ tree /opt/techstart/
/opt/techstart/
├── backups
├── logs [error opening dir]
├── projects
│   ├── backend
│   ├── frontend
│   └── infrastructure
├── shared
│   ├── docs
│   └── resources
└── 9 directories, 0 files
ubuntu@ip-172-31-16-184:~$ S
```

Step 4: Testing – Perform all 5 required tests

Step1: Backend can create files in backend directory

```

ubuntu@ip-172-31-16-184:~$ sudo su - backend1
backend1@ip-172-31-16-184:~$ cd /opt/techstart/projects/backend
backend1@ip-172-31-16-184:/opt/techstart/projects/backend$ touch test_file_from_backend1.txt
backend1@ip-172-31-16-184:/opt/techstart/projects/backend$ ls -l test_file_from_backend1.txt
-rw-r--r-- 1 backend1 backend-dev 0 Oct  6 14:40 test_file_from_backend1.txt
backend1@ip-172-31-16-184:/opt/techstart/projects/backend$ exit
logout
ubuntu@ip-172-31-16-184:~$ S

```

Step2: Frontend can't create files in Backend directory

```

ubuntu@ip-172-31-16-184:~$ sudo su - backend1
backend1@ip-172-31-16-184:~$ cd /opt/techstart/projects/backend
backend1@ip-172-31-16-184:/opt/techstart/projects/backend$ touch test_file_from_backend1.txt
backend1@ip-172-31-16-184:/opt/techstart/projects/backend$ ls -l test_file_from_backend1.txt
-rw-r--r-- 1 backend1 backend-dev 0 Oct  6 14:40 test_file_from_backend1.txt
backend1@ip-172-31-16-184:/opt/techstart/projects/backend$ exit
logout
ubuntu@ip-172-31-16-184:~$ sudo su - frontend1
frontend1@ip-172-31-16-184:~$ cd /opt/techstart/projects/backend
frontend1@ip-172-31-16-184:/opt/techstart/projects/backend$ touch test_file_from_frontend1.txt
touch: cannot touch 'test_file_from_frontend1.txt': Permission denied
frontend1@ip-172-31-16-184:/opt/techstart/projects/backend$ exit
logout
ubuntu@ip-172-31-16-184:~$ S

```

Test3: All developers can read and write in shared directory

Test with backend user

```

backend1@ip-172-31-16-184:~$ sudo su - backend1
[sudo] password for backend1:
backend1 is not in the sudoers file. This incident will be reported.
backend1@ip-172-31-16-184:~$ cd /opt/techstart/shared/
backend1@ip-172-31-16-184:/opt/techstart/shared$ touch shared_file_from_backend1.txt
backend1@ip-172-31-16-184:/opt/techstart/shared$ ls -l
total 8
drwxrwsr-x 2 root      dev-all 4096 Oct  6 14:28 docs
drwxrwsr-x 2 root      dev-all 4096 Oct  6 14:28 resources
-rw-r--r-- 1 backend1 dev-all  0 Oct  6 14:48 shared_file_from_backend1.txt
backend1@ip-172-31-16-184:/opt/techstart/shared$ S

```

#Test with frontend user

```

ubuntu@ip-172-31-16-184:~$ sudo su - frontend1
frontend1@ip-172-31-16-184:~$ cd /opt/techstart/shared/
frontend1@ip-172-31-16-184:/opt/techstart/shared$ touch shared_file_from_frontend1.txt
frontend1@ip-172-31-16-184:/opt/techstart/shared$ ls -l
total 8
drwxrwsr-x 2 root      dev-all 4096 Oct  6 14:28 docs
drwxrwsr-x 2 root      dev-all 4096 Oct  6 14:28 resources
-rw-r--r-- 1 backend1  dev-all   0 Oct  6 14:48 shared_file_from_backend1.txt
-rw-r--r-- 1 frontend1 dev-all   0 Oct  6 14:50 shared_file_from_frontend1.txt
frontend1@ip-172-31-16-184:/opt/techstart/shared$ S

```

#Test with devops user

```

logout
ubuntu@ip-172-31-16-184:~$ sudo su - devops1
devops1@ip-172-31-16-184:~$ cd /opt/techstart/shared/
devops1@ip-172-31-16-184:/opt/techstart/shared$ touch shared_file_from_devops1.txt
devops1@ip-172-31-16-184:/opt/techstart/shared$ ls -l
total 8
drwxrwsr-x 2 root      dev-all 4096 Oct  6 14:28 docs
drwxrwsr-x 2 root      dev-all 4096 Oct  6 14:28 resources
-rw-r--r-- 1 backend1  dev-all   0 Oct  6 14:48 shared_file_from_backend1.txt
-rw-r--r-- 1 devops1   dev-all   0 Oct  6 14:53 shared_file_from_devops1.txt
-rw-r--r-- 1 frontend1 dev-all   0 Oct  6 14:50 shared_file_from_frontend1.txt
devops1@ip-172-31-16-184:/opt/techstart/shared$ S

```

Test4: Only team lead can write to logs directory

#Test with regular developer (Should fail)

```

ubuntu@ip-172-31-16-184:~$ sudo su - backend1
backend1@ip-172-31-16-184:~$ cd /opt/techstart/logs/
-bash: cd: /opt/techstart/logs/: Permission denied
backend1@ip-172-31-16-184:~$ ^C

```

#Test with team lead (should succeed)

```

ubuntu@ip-172-31-16-184:~$ sudo su - teamlead
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

teamlead@ip-172-31-16-184:~$ cd /opt/techstart/logs/
teamlead@ip-172-31-16-184:/opt/techstart/logs$ touch log_file_from_teamlead.txt
teamlead@ip-172-31-16-184:/opt/techstart/logs$ ls -l
total 0
-rw-r--r-- 1 teamlead team-leads 0 Oct  6 15:04 log_file_from_teamlead.txt
teamlead@ip-172-31-16-184:/opt/techstart/logs$

```

Test 5: Verify SGID is working – new files inherit directory's group

#Check multiple users creating files in their respective directories

```
ubuntu@ip-172-31-16-184:~$ sudo su - backend1
backend1@ip-172-31-16-184:~$ cd /opt/techstart/projects/backend
backend1@ip-172-31-16-184:/opt/techstart/projects/backend$ touch backend_test_sg
id.txt
backend1@ip-172-31-16-184:/opt/techstart/projects/backend$ ls -l backend_test_sg
id.txt
-rw-r--r-- 1 backend1 backend-dev 0 Oct  6 15:09 backend_test_sgid.txt
backend1@ip-172-31-16-184:/opt/techstart/projects/backend$ S
```

```
ubuntu@ip-172-31-16-184:~$ sudo su - frontend1
frontend1@ip-172-31-16-184:~$ cd /opt/techstart/projects/frontend
frontend1@ip-172-31-16-184:/opt/techstart/projects/frontend$ touch frontend_test
_sgid.txt
frontend1@ip-172-31-16-184:/opt/techstart/projects/frontend$ ls -l frontend_test
_sgid.txt
-rw-r--r-- 1 frontend1 frontend-dev 0 Oct  6 15:11 frontend_test_sgid.txt
frontend1@ip-172-31-16-184:/opt/techstart/projects/frontend$ exit
logout
ubuntu@ip-172-31-16-184:~$ sudo su - devops1
devops1@ip-172-31-16-184:~$ cd /opt/techstart/projects/infrastructure
devops1@ip-172-31-16-184:/opt/techstart/projects/infrastructure$ touch devops_te
st_sgid.txt
devops1@ip-172-31-16-184:/opt/techstart/projects/infrastructure$ ls -l devops_te
st_sgid.txt
-rw-r--r-- 1 devops1 devops-team 0 Oct  6 15:13 devops_test_sgid.txt
devops1@ip-172-31-16-184:/opt/techstart/projects/infrastructure$ S
```

Step 5: Documentation Commands

```
ubuntu@ip-172-31-16-184:~$ tree /opt/techstart/
/opt/techstart/
├── backups
├── logs [error opening dir]
├── projects
│   ├── backend
│   │   ├── backend_test_sgid.txt
│   │   └── test_file_from_backend1.txt
│   ├── frontend
│   │   └── frontend_test_sgid.txt
│   └── infrastructure
│       └── devops_test_sgid.txt
├── shared
│   ├── docs
│   ├── resources
│   ├── shared_file_from_backend1.txt
│   ├── shared_file_from_devops1.txt
│   └── shared_file_from_frontend1.txt
9 directories, 7 files
ubuntu@ip-172-31-16-184:~$ S
```

```

drwxr-xr-x 5 root root      4096 Oct  6 14:28 projects
drwxrwsr-x 4 root dev-all  4096 Oct  6 14:53 shared
ubuntu@ip-172-31-16-184:~$ ls -la /opt/techstart/projects/
total 20
drwxr-xr-x 5 root root      4096 Oct  6 14:28 .
drwxr-xr-x 6 root root      4096 Oct  6 14:28 ..
drwxrwsr-x 2 root backend-dev 4096 Oct  6 15:09 backend
drwxrwsr-x 2 root frontend-dev 4096 Oct  6 15:11 frontend
drwxrwsr-x 2 root devops-team 4096 Oct  6 15:13 infrastructure
ubuntu@ip-172-31-16-184:~$ ls -la /opt/techstart/shared/
total 16
drwxrwsr-x 4 root      dev-all 4096 Oct  6 14:53 .
drwxr-xr-x 6 root      root     4096 Oct  6 14:28 ..
drwxrwsr-x 2 root      dev-all 4096 Oct  6 14:28 docs
drwxrwsr-x 2 root      dev-all 4096 Oct  6 14:28 resources
-rw-r--r-- 1 backend1  dev-all   0 Oct  6 14:48 shared_file_from_backend1.txt
-rw-r--r-- 1 devops1   dev-all   0 Oct  6 14:53 shared_file_from_devops1.txt
-rw-r--r-- 1 frontend1 dev-all   0 Oct  6 14:50 shared_file_from_frontend1.txt
ubuntu@ip-172-31-16-184:~$ ls -la /opt/techstart/logs/
ls: cannot open directory '/opt/techstart/logs/': Permission denied
ubuntu@ip-172-31-16-184:~$ ls -la /opt/techstart/projects/backend | grep "^d"
drwxrwsr-x 2 root      backend-dev 4096 Oct  6 15:09 .
drwxr-xr-x 5 root      root         4096 Oct  6 14:28 ..
ubuntu@ip-172-31-16-184:~$

```

PART 5: Sudo Configuration – Complete Step-by-step instructions

Step 1: Create Sudo configuration file

Critical: Always use visudo for sudo configuration

Step 2: Add the following lines to the file

```

GNU nano 6.2 /etc/sudoers.d/techstart.tmp *
# Team leads have full access
%team-leads    ALL=(ALL:ALL) ALL

# DevOps can manage services without a password
%devops-team   ALL=(ALL:ALL) NOPASSWD: /bin/systemctl restart *, /bin/systemctl
# DevOps can read logs (but require password)
</log/*

```

STEP 3: Verification – Check the file was created

STEP 4: Testing – Perform All 4 Required Tests

Test 1: Team lead has full sudo access

```

ubuntu@ip-172-31-16-184:~$ sudo su - teamlead
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

teamlead@ip-172-31-16-184:~$ sudo -i
[sudo] password for teamlead:
root@ip-172-31-16-184:~# exit
logout
teamlead@ip-172-31-16-184:~$ sudo whoami
root
teamlead@ip-172-31-16-184:~$ █

```

Test 2: Devops user can restart services without password

```

teamlead@ip-172-31-16-184:~$ sudo su - devops1
devops1@ip-172-31-16-184:~$ sudo systemctl restart ssh
devops1@ip-172-31-16-184:~$ sudo systemctl stop ssh
devops1@ip-172-31-16-184:~$ sudo systemctl start ssh
devops1@ip-172-31-16-184:~$ exit
logout

```

Test 3: Devops user needs password for reading logs

Command used: `sudo tail /var/log/syslog`

It shows the last 10 lines of the system log file in real-time.

```

ubuntu@ip-172-31-16-184:~$ sudo su - devops1
devops1@ip-172-31-16-184:~$ sudo less /var/log/syslog
[sudo] password for devops1:

[1]+  Stopped                  sudo less /var/log/syslog
devops1@ip-172-31-16-184:~$ sudo tail /var/log/syslog
Oct  6 15:41:36 ip-172-31-16-184 systemd[1]: Stopping OpenBSD Secure Shell server...
Oct  6 15:41:36 ip-172-31-16-184 systemd[1]: ssh.service: Deactivated successfully.
Oct  6 15:41:36 ip-172-31-16-184 systemd[1]: Stopped OpenBSD Secure Shell server.
Oct  6 15:41:36 ip-172-31-16-184 systemd[1]: Starting OpenBSD Secure Shell server...
Oct  6 15:41:36 ip-172-31-16-184 systemd[1]: Started OpenBSD Secure Shell server.
Oct  6 15:41:49 ip-172-31-16-184 systemd[1]: Stopping OpenBSD Secure Shell server...
Oct  6 15:41:49 ip-172-31-16-184 systemd[1]: ssh.service: Deactivated successfully.

```

Test 4: Regular developer has no sudo access

This Is to prove that regular users are limited to some access.

```
devops1@ip-172-31-16-184:~$ sudo su - backend1
Sorry, user devops1 is not allowed to execute '/usr/bin/su - backend1' as root o
n ip-172-31-16-184.eu-north-1.compute.internal.
devops1@ip-172-31-16-184:~$ sudo ls
Sorry, user devops1 is not allowed to execute '/usr/bin/ls' as root on ip-172-31
-16-184.eu-north-1.compute.internal.
devops1@ip-172-31-16-184:~$
```

STEP 5: Additional Verification Tests

Command used: `sudo systemctl restart ssh`

- `sudo`: Execute with administrator privileges
- `systemctl`: System control utility for managing services
- `restart`: Stop and then start a service
- `ssh`: The SSH daemon/service that allows remote connections

SSH (Secure Shell) - A network protocol for securely connecting to and managing remote computers over an unsecured network. It provides encrypted communication between client and server.

```
ubuntu@ip-172-31-16-184:~$ sudo su - devops2
devops2@ip-172-31-16-184:~$ sudo systemctl restart ssh
devops2@ip-172-31-16-184:~$ sudo less /var/log/syslog
[sudo] password for devops2:

[1]+  Stopped                  sudo less /var/log/syslog
devops2@ip-172-31-16-184:~$ exit
logout
There are stopped jobs.
devops2@ip-172-31-16-184:~$ exit
logout
ubuntu@ip-172-31-16-184:~$ sudo su - devops3
devops3@ip-172-31-16-184:~$ sudo systemctl restart ssh
devops3@ip-172-31-16-184:~$ sudo less /var/log/syslog
[sudo] password for devops3:

[1]+  Stopped                  sudo less /var/log/syslog
devops3@ip-172-31-16-184:~$ exit
logout
There are stopped jobs.
devops3@ip-172-31-16-184:~$ exit
logout
sudo: mon_handle_sigchld: waitpid: No child processes
ubuntu@ip-172-31-16-184:~$
```

STEP 6: Documentation Commands

```

sudo: Mon_handle_stgcmd: waitpid: no child processes
ubuntu@ip-172-31-16-184:~$ sudo cat /etc/sudoers.d/techstart
# Team leads have full access
%team-leads    ALL=(ALL:ALL) ALL

# DevOps can manage services without a password
%devops-team   ALL=(ALL:ALL) NOPASSWD: /bin/systemctl restart *, /bin/systemctl
stop *, /bin/systemctl start *

# DevOps can read logs (but require password)
%devops-team   ALL=(ALL:ALL) /usr/bin/less /var/log/*, /usr/bin/tail /var/log/*
ubuntu@ip-172-31-16-184:~$ sudo visudo -c
/etc/sudoers: parsed OK
/etc/sudoers.d/90-cloud-init-users: parsed OK
/etc/sudoers.d/README: parsed OK
/etc/sudoers.d/techstart: parsed OK
ubuntu@ip-172-31-16-184:~$ sudo -l -U teamlead
Matching Defaults entries for teamlead on ip-172-31-16-184:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bi
n\:/snap/bin,
    use_pty

```

Command used: `sudo -l -U devops1`

- `sudo`: Execute with elevated privileges
- `-l`: List mode (show privileges)
- `-U devops1`: Specify the user to check (devops1)

```

ubuntu@ip-172-31-16-184:~$ sudo -l -U devops1
Matching Defaults entries for devops1 on ip-172-31-16-184:
    env_reset, mail_badpass,
    secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bi
n\:/snap/bin,
    use_pty

User devops1 may run the following commands on ip-172-31-16-184:
    (ALL : ALL) NOPASSWD: /bin/systemctl restart *, /bin/systemctl stop *,
    /bin/systemctl start *
    (ALL : ALL) /usr/bin/less /var/log/*, /usr/bin/tail /var/log/*
ubuntu@ip-172-31-16-184:~$ sudo -l -U backend1
User backend1 is not allowed to run sudo on ip-172-31-16-184.
ubuntu@ip-172-31-16-184:~$ S

```

PART 6: Password Security Policies

Step 1: Set Password policies for all 10 users

Full Command Breakdown:

`sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 backend1`

- **chage:** Change user password aging/expiry settings
- **M 90:** Maximum password age = 90 days
- **m 7:** Minimum password age = 7 days (can't change before 7 days)
- **W 14:** Warning period = 14 days before expiry
- **I 30:** Inactive period = 30 days after expiry before account locks
- **E -1:** Account expiration date = never
- **backend1:** The username being configured

```
ubuntu@ip-172-31-16-184:~$ sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 backend1
sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 backend2
sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 backend3
ubuntu@ip-172-31-16-184:~$ sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 frontend1
sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 frontend2
sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 frontend3
ubuntu@ip-172-31-16-184:~$ sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 frontend3
ubuntu@ip-172-31-16-184:~$ sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 devops1
sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 devops2
sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 devops3
ubuntu@ip-172-31-16-184:~$ sudo chage -M 90 -m 7 -W 14 -I 30 -E -1 teamlead
ubuntu@ip-172-31-16-184:~$ SS
```

Step 2: Verification- Check Password Settings

```
Last password change           : Oct 06, 2025
Password expires               : Jan 04, 2026
Password inactive              : Feb 03, 2026
Account expires                : never
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14
ubuntu@ip-172-31-16-184:~$ sudo chage -l frontend1
Last password change           : Oct 06, 2025
Password expires               : Jan 04, 2026
Password inactive              : Feb 03, 2026
Account expires                : never
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14
ubuntu@ip-172-31-16-184:~$ sudo chage -l devops1
Last password change           : Oct 06, 2025
Password expires               : Jan 04, 2026
Password inactive              : Feb 03, 2026
Account expires                : never
Minimum number of days between password change : 7
Maximum number of days between password change : 90
Number of days of warning before password expires : 14
ubuntu@ip-172-31-16-184:~$
```

PART 7: User Information and Monitoring

STEP1: User Inventory – List all regular users

```
ubuntu@ip-172-31-16-184:~$ getent passwd | awk -F: '$3 >= 1000 {print $1, $3, $4}' | column -t
nobody                65534    65534
ubuntu                1000     1000
sano_kabayiza_jules   1001     1001
iradukunda_oscar      1002     1002
ishimwe_amos          1003     1003
backend1              1004     1004
backend2              1005     1004
backend3              1006     1004
frontend1             1007     1005
frontend2             1008     1005
frontend3             1009     1005
devops1               1010     1006
devops2               1011     1006
devops3               1012     1006
teamlead              1013     1008
ubuntu@ip-172-31-16-184:~$
```

STEP2: Group Membership Report

```
ubuntu@ip-172-31-16-184:~$ getent group backend-dev frontend-dev devops-team dev
-all team-leads sudo
backend-dev:x:1004:
frontend-dev:x:1005:
devops-team:x:1006:
dev-all:x:1007:backend1,backend2,backend3,frontend1,frontend2,frontend3,devops1,
devops2,devops3,teamlead
team-leads:x:1008:
sudo:x:27:ubuntu,sano_kabayiza_jules,iradukunda_oscar,ishimwe_amos,teamlead
ubuntu@ip-172-31-16-184:~$ for group in backend-dev frontend-dev devops-team dev
-all team-leads; do
    echo "Group: $group (GID: $(getent group $group | cut -d: -f3))"
    getent group $group | cut -d: -f4
    echo
done
Group: backend-dev (GID: 1004)

Group: frontend-dev (GID: 1005)

Group: devops-team (GID: 1006)
```

```

ubuntu@ip-172-31-16-184:~$ for group in backend-dev frontend-dev devops-team dev
-all team-leads; do
    echo "Group: $group (GID: $(getent group $group | cut -d: -f3))"
    getent group $group | cut -d: -f4
    echo
done
Group: backend-dev (GID: 1004)

Group: frontend-dev (GID: 1005)

Group: devops-team (GID: 1006)

Group: dev-all (GID: 1007)
backend1,backend2,backend3,frontend1,frontend2,frontend3,devops1,devops2,devops3
,teamlead

Group: team-leads (GID: 1008)

ubuntu@ip-172-31-16-184:~$

```

STEP3: Sudo Access Report

```

ubuntu@ip-172-31-16-184:~$ getent group sudo
sudo:x:27:ubuntu,sano_kabayiza_jules,iradukunda_oscar,ishimwe_amos,teamlead
ubuntu@ip-172-31-16-184:~$ sudo cat /etc/sudoers.d/techstart
# Team leads have full access
%team-leads    ALL=(ALL:ALL) ALL

# DevOps can manage services without a password
%devops-team   ALL=(ALL:ALL) NOPASSWD: /bin/systemctl restart *, /bin/systemctl
stop *, /bin/systemctl start *

# DevOps can read logs (but require password)
%devops-team   ALL=(ALL:ALL) /usr/bin/less /var/log/*, /usr/bin/tail /var/log/*
ubuntu@ip-172-31-16-184:~$ for user in teamlead devops1 backend1; do
    echo "--- $user sudo privileges ---"
    sudo -l -U $user 2>/dev/null | grep -E "(User|may run|NOPASSWD)" || echo "No
sudo access"
    echo
done
--- teamlead sudo privileges ---
User teamlead may run the following commands on ip-172-31-16-184:

--- devops1 sudo privileges ---
User devops1 may run the following commands on ip-172-31-16-184:

```

STEP4: Resource Usage Report

```

ubuntu@ip-172-31-16-184:~$ sudo du -sh /home/*
20K    /home/backend1
16K    /home/backend2
16K    /home/backend3
20K    /home/devops1
20K    /home/devops2
20K    /home/devops3
20K    /home/frontend1
16K    /home/frontend2
16K    /home/frontend3
16K    /home/iradukunda_oscar
16K    /home/ishinwe_amos
16K    /home/sano_kabayiza_jules
20K    /home/teamlead
32K    /home/ubuntu
ubuntu@ip-172-31-16-184:~$ sudo du -sh /opt/techstart/
40K    /opt/techstart/
ubuntu@ip-172-31-16-184:~$ sudo du -sh /opt/techstart/*
4.0K   /opt/techstart/backups
4.0K   /opt/techstart/logs
16K    /opt/techstart/projects
12K    /opt/techstart/shared
ubuntu@ip-172-31-16-184:~$

```

STEP5: Login Activity Report

```

ubuntu@ip-172-31-16-184:~$ lastlog
Username      Port      From      Latest
root          0         0.0.0.0    **Never logged in**
daemon        0         0.0.0.0    **Never logged in**
bin           0         0.0.0.0    **Never logged in**
sys          0         0.0.0.0    **Never logged in**
sync         0         0.0.0.0    **Never logged in**
games        0         0.0.0.0    **Never logged in**
man          0         0.0.0.0    **Never logged in**
lp           0         0.0.0.0    **Never logged in**
mail         0         0.0.0.0    **Never logged in**
news         0         0.0.0.0    **Never logged in**
uucp         0         0.0.0.0    **Never logged in**
proxy        0         0.0.0.0    **Never logged in**
www-data     0         0.0.0.0    **Never logged in**
backup       0         0.0.0.0    **Never logged in**
list         0         0.0.0.0    **Never logged in**
irc          0         0.0.0.0    **Never logged in**

```

```

ubuntu@ip-172-31-16-184:~$ who
ubuntu pts/1      2025-10-06 12:48 (102.22.168.153)
ubuntu pts/2      2025-10-06 16:10 (102.22.168.153)
ubuntu pts/3      2025-10-06 14:49 (102.22.168.153)
ubuntu pts/4      2025-10-06 15:12 (102.22.168.153)
ubuntu pts/5      2025-10-06 15:20 (102.22.168.153)
ubuntu pts/6      2025-10-06 15:31 (102.22.168.153)
ubuntu pts/7      2025-10-06 16:42 (102.22.168.153)
ubuntu@ip-172-31-16-184:~$ last -10
ubuntu pts/7      102.22.168.153  Mon Oct  6 16:42  still logged in
ubuntu pts/2      102.22.168.153  Mon Oct  6 16:10  still logged in
ubuntu pts/6      102.22.168.153  Mon Oct  6 15:31  still logged in
ubuntu pts/5      102.22.168.153  Mon Oct  6 15:20  still logged in
ubuntu pts/3      102.22.168.153  Mon Oct  6 14:49  still logged in
ubuntu pts/3      102.22.168.153  Mon Oct  6 14:06 - 14:49  (00:42)
ubuntu pts/2      102.22.168.153  Mon Oct  6 13:10 - 15:31  (02:21)
ubuntu pts/0      102.22.168.153  Mon Oct  6 12:48 - 15:13  (02:25)
reboot system boot  6.8.0-1035-aws  Mon Oct  6 11:02  still running

wtmp begins Mon Oct  6 11:02:19 2025
ubuntu@ip-172-31-16-184:~$ ss

```

STEP6: File Ownership Analysis

```

ubuntu@ip-172-31-16-184:~$ for user in backend1 backend2 backend3 frontend1 frontend2 frontend3 devops1 devops2 devops3 teamlead ubuntu; do
    count=$(sudo find / -type f -user $user 2>/dev/null | wc -l)
    echo "$user: $count files"
done
backend1: 7 files
backend2: 3 files
backend3: 3 files
frontend1: 6 files
frontend2: 3 files
frontend3: 3 files
devops1: 123 files
devops2: 4 files
devops3: 4 files
teamlead: 6 files
ubuntu: 1121 files
ubuntu@ip-172-31-16-184:~$ for user in backend1 frontend1 devops1; do
    echo "--- $user files outside /home/$user ---"
    sudo find / -type f -user $user 2>/dev/null | grep -v "^/home/$user" | head
-5
    echo
done
--- backend1 files outside /home/backend1 ---
/opt/techstart/shared/shared file from backend1.txt

```

```

ubuntu@ip-172-31-16-184:~$ for user in backend1 frontend1 devops1; do
    echo "--- $user files outside /home/$user ---"
    sudo find / -type f -user $user 2>/dev/null | grep -v "^/home/$user" | head
-5
    echo
done
--- backend1 files outside /home/backend1 ---
/opt/techstart/shared/shared_file_from_backend1.txt
/opt/techstart/projects/backend/backend_test_sgid.txt
/opt/techstart/projects/backend/test_file_from_backend1.txt

--- frontend1 files outside /home/frontend1 ---
/opt/techstart/shared/shared_file_from_frontend1.txt
/opt/techstart/projects/frontend/frontend_test_sgid.txt

--- devops1 files outside /home/devops1 ---
/proc/9863/task/9863/fdinfo/0
/proc/9863/task/9863/fdinfo/1
/proc/9863/task/9863/fdinfo/2
/proc/9863/task/9863/fdinfo/255
/proc/9863/task/9863/environ
ubuntu@ip-172-31-16-184:~$ S

```

STEP 7: COMPREHENSIVE SYSTEM REPORT

```

SYSTEM OVERVIEW:
Hostname: ip-172-31-16-184
Users created: 15
Groups created: 5

USER SUMMARY TABLE:
Username | UID | Primary Group | Home Directory | Shell
-----|-----|-----|-----|-----
nobody | 65534 | nogroup | /nonexistent | /usr/sbin/nologin
sano_kabayiza_jules | 1001 | sano_kabayiza_jules | /home/sano_kabayiza_jules | /bin/bash
iradukunda_oscar | 1002 | iradukunda_oscar | /home/iradukunda_oscar | /bin/bash
ishimwe_amos | 1003 | ishimwe_amos | /home/ishimwe_amos | /bin/bash
backend1 | 1004 | backend-dev | /home/backend1 | /bin/bash
backend2 | 1005 | backend-dev | /home/backend2 | /bin/bash
backend3 | 1006 | backend-dev | /home/backend3 | /bin/bash
frontend1 | 1007 | frontend-dev | /home/frontend1 | /bin/bash
frontend2 | 1008 | frontend-dev | /home/frontend2 | /bin/bash
frontend3 | 1009 | frontend-dev | /home/frontend3 | /bin/bash
devops1 | 1010 | devops-team | /home/devops1 | /bin/bash
devops2 | 1011 | devops-team | /home/devops2 | /bin/bash
devops3 | 1012 | devops-team | /home/devops3 | /bin/bash
teamlead | 1013 | team-leads | /home/teamlead | /bin/bash
ubuntu@ip-172-31-16-184:~$ S

```

STEP 8: CREATE ORGANIZED DOCUMENTATION TABLES

```

=== TABLE 1: USER INVENTORY ===
Username | UID | Primary Group ID
-----|-----|-----
nobody | 65534 | 65534
sano_kabayiza_jules | 1001 | 1001
iradukunda_oscar | 1002 | 1002
ishimwe_amos | 1003 | 1003
backend1 | 1004 | 1004
backend2 | 1005 | 1004
backend3 | 1006 | 1004
frontend1 | 1007 | 1005
frontend2 | 1008 | 1005
frontend3 | 1009 | 1005
devops1 | 1010 | 1006
devops2 | 1011 | 1006
devops3 | 1012 | 1006
teamlead | 1013 | 1008

=== TABLE 2: GROUP MEMBERSHIP MATRIX ===
Group Name | GID | Members
-----|-----|-----
backend-dev | 1004 |
frontend-dev | 1005 |

```

```

=== TABLE 2: GROUP MEMBERSHIP MATRIX ===
Group Name | GID | Members
-----|-----|-----
backend-dev | 1004 |
frontend-dev | 1005 |
devops-team | 1006 |
dev-all | 1007 | backend1,backend2,backend3,frontend1,frontend2,frontend3,devops
1,devops2,devops3,teamlead
team-leads | 1008 |

=== TABLE 3: SUDO ACCESS REPORT ===
User/Group | Access Level | Password Required
-----|-----|-----
team-leads | Full sudo access | Yes
devops-team | Service management commands | No (for services)
devops-team | Log reading commands | Yes
Regular developers | No sudo access | N/A

```

```

=== TABLE 4: DISK USAGE REPORT ===
Directory | Size
-----|-----
/home/backend1 | 20K
/home/frontend1 | 20K
/home/devops1 | 20K
/home/teamlead | 20K
/opt/techstart/ | 40K

=== TABLE 5: LOGIN ACTIVITY SUMMARY ===
Username | Last Login
-----|-----
Username |
ubuntu | Oct 6 16:42:05 +0000
ubuntu@ip-172-31-16-184:~$ ss

```

PART 8: User Removal

Phase 1: Account Suspension

#Lock user account to prevent login

```
ubuntu@ip-172-31-16-184:~$ sudo usermod -L frontend3
ubuntu@ip-172-31-16-184:~$ sudo passwd -S frontend3
frontend3 L 10/06/2025 7 90 14 30
ubuntu@ip-172-31-16-184:~$ sudo su - frontend3
frontend3@ip-172-31-16-184:~$ S
```

Phase 2: Data Backup

```
ubuntu@ip-172-31-16-184:~$ sudo mkdir -p /opt/techstart/backups
ubuntu@ip-172-31-16-184:~$ sudo tar -czf /opt/techstart/backups/frontend3_backup_
$(date +%Y%m%d).tar.gz /home/frontend3
tar: Removing leading '/' from member names
ubuntu@ip-172-31-16-184:~$ ls -lh /opt/techstart/backups/
total 4.0K
-rw-r--r-- 1 root root 2.3K Oct  6 17:20 frontend3_backup_20251006.tar.gz
ubuntu@ip-172-31-16-184:~$ sudo ls -la /opt/techstart/backups/frontend3_backup_*.tar.gz
-rw-r--r-- 1 root root 2346 Oct  6 17:20 /opt/techstart/backups/frontend3_backup_20251006.tar.gz
ubuntu@ip-172-31-16-184:~$ sudo tar -tzf /opt/techstart/backups/frontend3_backup_*.tar.gz | head -10
home/frontend3/
home/frontend3/.bash_history
home/frontend3/.bash_logout
home/frontend3/.profile
home/frontend3/.bashrc
ubuntu@ip-172-31-16-184:~$
```

Phase 3: File Discovery

```
ubuntu@ip-172-31-16-184:~$ sudo find / -user frontend3 2>/dev/null
/home/frontend3
/home/frontend3/.bash_history
/home/frontend3/.bash_logout
/home/frontend3/.profile
/home/frontend3/.bashrc
ubuntu@ip-172-31-16-184:~$ sudo find / -user frontend3 2>/dev/null | wc -l
5
ubuntu@ip-172-31-16-184:~$ sudo find / -user frontend3 2>/dev/null | grep -v "^/home/frontend3"
ubuntu@ip-172-31-16-184:~$ echo "=== FILES OWNED BY frontend3 ==="
sudo find / -user frontend3 2>/dev/null | while read file; do
    echo "File: $file"
    ls -la "$file" 2>/dev/null | head -1
done
=== FILES OWNED BY frontend3 ===
File: /home/frontend3
File: /home/frontend3/.bash_history
File: /home/frontend3/.bash_logout
File: /home/frontend3/.profile
File: /home/frontend3/.bashrc
ubuntu@ip-172-31-16-184:~$ S
```

Phase 4: Account Deletion

```
ubuntu@ip-172-31-16-184:~$ sudo userdel -r frontend3
userdel: frontend3 mail spool (/var/mail/frontend3) not found
ubuntu@ip-172-31-16-184:~$ echo "User deletion completed at: $(date)"
User deletion completed at: Mon Oct 6 17:25:43 UTC 2025
ubuntu@ip-172-31-16-184:~$ S
```

Phase 5: Verification

```
echo "=== VERIFICATION STEP 5: Home directory check ==="
ls -la /home/frontend3 2>/dev/null || echo "Home directory /home/frontend3 does not exist"
ls -la /opt/techstart/backups/frontend3_backup_*.tar.gz
=== VERIFICATION STEP 1: User existence check ===
=== VERIFICATION STEP 2: Group membership check ===
frontend-dev:x:1005:
dev-all:x:1007:backend1,backend2,backend3,frontend1,frontend2,devops1,devops2,devops3,teamlead
=== VERIFICATION STEP 3: Process check ===
pgrep: invalid user name: frontend3
=== VERIFICATION STEP 4: File ownership check ===
=== VERIFICATION STEP 5: Home directory check ===
Home directory /home/frontend3 does not exist (as expected)
=== VERIFICATION STEP 6: Backup verification ===
-rw-r--r-- 1 root root 2346 Oct 6 17:20 /opt/techstart/backups/frontend3_backup_20251006.tar.gz
ubuntu@ip-172-31-16-184:~$ S
```

Additional Verification Steps

```
# Check system logs for any issues
echo "=== SYSTEM LOG CHECK ==="
sudo tail -5 /var/log/auth.log | grep frontend3 || echo "No recent auth log entries for frontend3"
=== ADDITIONAL VERIFICATION: Authentication test ===
User frontend3 does not exist
=== GROUP MEMBERSHIP VERIFICATION ===
Group frontend-dev members:
Group dev-all members: backend1,backend2,backend3,frontend1,frontend2,devops1,devops2,devops3,teamlead
=== SYSTEM LOG CHECK ===
Oct 6 17:26:58 ip-172-31-16-184 sudo: ubuntu : TTY=pts/8 ; PWD=/home/ubuntu ; USER=root ; COMMAND=/usr/bin/find / -user frontend3
ubuntu@ip-172-31-16-184:~$
```

Complete Documentation Commands

```
=== USER REMOVAL COMPLETE REPORT ===
User: frontend3 (Emily Davis)
Removal Date: Mon Oct 6 17:29:56 UTC 2025

PHASE 1 - ACCOUNT SUSPENSION:
Account no longer exists (expected)

PHASE 2 - BACKUP CREATION:
-rw-r--r-- 1 root root 2346 Oct 6 17:20 /opt/techstart/backups/frontend3_backup_20251006.tar.gz

PHASE 3 - FILE DISCOVERY (BEFORE DELETION):
Files were documented during Phase 3

PHASE 4 - ACCOUNT DELETION:
User account and home directory removed with: sudo userdel -r frontend3

PHASE 5 - VERIFICATION:
1. User exists in passwd: NO
2. User in frontend-dev group: NO
3. User in dev-all group: NO
pgrep: invalid user name: frontend3
```

Conclusion:

Key Learning Outcomes

1. User and Group Management

- Learned to create and manage user groups using `groupadd`
- Mastered user creation with specific group assignments using `useradd`
- Understood the importance of proper group membership for collaboration

2. File System Permissions

- Gained hands-on experience with Linux file permissions
- Learned to navigate permission restrictions and solutions
- Understood how to verify file existence and ownership

3. Command Line Proficiency

- `sudo groupadd testing` - Creating system groups
- `sudo useradd -m -g testing testing1` - User creation with home directories and group assignment
- `ls -la` - Comprehensive directory listing with hidden files and detailed permissions
- `getent group` and `groups` - System information queries

4. Troubleshooting Skills

- Encountered and resolved real-world permission issues
- Learned when and why to use sudo privileges
- Developed problem-solving approaches for access denied scenarios

Technical Skills Demonstrated

System Administration

- Creating and managing user accounts
- Group-based access control
- Home directory management
- System file modification permissions

Security Understanding

- Principle of least privilege
- Proper use of sudo for system changes
- File permission best practices
- User isolation in multi-user environments

Verification Techniques

- Multiple methods to confirm group membership
- File existence and ownership verification
- Comprehensive system state checking

Key Insights

1. Order of operations matters - Groups must exist before assigning users
2. Permission hierarchy is crucial - Regular users vs. root privileges
3. Verification is essential - Always confirm changes took effect
4. Documentation through commands - Self-documenting verification processes

Project Success Metrics

- Group 'testing' created successfully
- Users 'testing1' and 'testing2' added to the group
- File 'test.txt' created in testing1's home directory

- All components properly verified
- Permission challenges identified and resolved

Educational Value

This project provided practical experience in fundamental Linux system administration tasks that form the foundation for more advanced topics like:

- Advanced permission models (ACLs)
- Automated user provisioning
- Security hardening
- Multi-user environment management

The hands-on approach ensured deep understanding of both the "how" and "why" behind Linux user and group management systems.

Project Status: COMPLETED SUCCESSFULLY